# Analysis and Synthesis of an Ant Colony Optimization Technique for Image Edge Detection

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***Abstract*: - Ant Colony optimization (ACO) is the technique which is used for solving computational problems and finding the best paths through graphs. ACO is based on the behavior ofants seeking paths from their colony to their food. Ants move randomly and after getting their food return back to their colony while laying down pheromone trails. Other ants find such a path and follow trail for returning. Pheromones are used for ant's communication. This technique is used for optimization in many applications like edge detection, network packet routing, structure health monitoring, vehicular routing, image segmentation traveling salesman problem, quadratic assignment problem, sequential ordering, scheduling, graph coloring, management of communications networks, image compression etc. In this paper we are using a method using ACO to find edge detection. It gives a pheromone matrix and memory stored positions that are followed by leading ant. The memory based positions are stored on the basis of intensity values with reference with a threshold value. The results are shown which successfully detect the edges of the image.**

***Keywords: - ACO, Edge detection, Threshold Value, Peak Signal Noise Ratio (PSNR), Sobel Edge Detection, Canny Edge Detection.***

## I. INTRODUCTION

Ant colony optimization is nature inspired optimization technique that is based on the general behavior of the ants i.e. how the ants wander randomly from source to food [2]. Ants deposit pheromone to the ground to mark their paths, which are followed by other ants, and over time pheromone evaporate. No. of pheromone on shorter paths are more because pheromone lay down faster by ants. This mechanism results in selecting shorter path. ACO is meta-heuristic approach [4]. The first ACO algorithm, called the *ant system*, was proposed by Dorigo *et al.*

The ACO is applied to many problems, in this paper; ACO is applied for Edge Detection. Edge detection is the process of extracting the edge information from the image so it is decisive to understand the image's content. In the proposed technique, the number of ant's moves on the image at pixel level and where there is variation of image intensity value as referred to some threshold value it store the position in memory storage and update the pheromone matrix [3]. The threshold value is described to represents the edge information at each pixel location of the image. In this proposed work the edge information is stored in the memory storage as on processing time and the results are found simultaneously. The technique proposed in this paper also work on different threshold values. The details of the technique are covered in this paper.

## II. IMAGE EDGE DETECTION

Edge detection is a fundamental tool in image processing and computer vision, particularly in the areas of feature detection and feature extraction, which aim at identifying points in a digital image at which the image brightness changes sharply or more formally has discontinuities [5]. The purpose of detecting sharp changes in image brightness is to capture important events and changes in properties of the world. It can be shown that under rather general assumptions for an image formation model, discontinuities in image brightness are likely to correspond to

- Discontinuities in depth,
- Discontinuities in surface orientation,
- Changes in material properties and
- Variations in scene illumination.

In the ideal case, the result of applying an edge detector to an image may lead to a set of connected curves that indicate the boundaries of objects, the boundaries of surface markings as well as curves that correspond to discontinuities in surface orientation [5]. Thus, applying an edge detection algorithm to an image may significantly reduce the amount of data to be processed and may therefore filter out information that may be regarded as less relevant, while preserving the important structural properties of an image. If the edge detection step is successful, the subsequent task of interpreting the information contents in the original image may therefore be substantially simplified [6]. However, it is not always possible to obtain such ideal edges from real life images of moderate complexity. Edges extracted from non-trivial images are often hampered by fragmentation, meaning that the edge curves are not connected, missing edge segments as well as false edges not corresponding to interesting phenomena in the image – thus complicating the subsequent task of interpreting the image data [4]. Edge detection is one of the fundamental steps in image processing, image analysis, image pattern recognition, and computer vision techniques.

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Image edge detection is the process of extracting the edges of digital images [4]. It is helpful in understand the content of an image for different applications. An ACO-based approach is used to overcome the limitations of conventional approaches. This paper gives a proposed approach of ACO in which decisive data is taken for further processing.

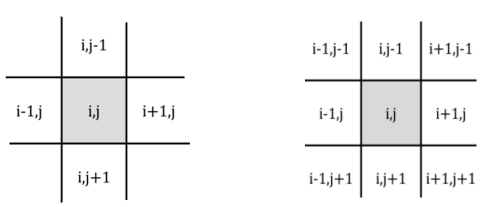
## III. ANT COLONY OPTIMIZATION

ACO mainly results to find optimal solution iteratively. The main work is to update the pheromone information which is given by the leading ants. If K ants are applied to the image it gives K optimal solutions. Let n iterations are made to store the memory position. The Proposed algorithm given is

**ALGORITHM [edge detection]**

**do**
initialization [Define initial points and Pheromone Matrix
$\boldsymbol{\tau}(0)$.]
**for each** iteration n=1:N
**do**
**for each** positions construction m=1:M such that
M<= N;
**do**
**for each** ant k=1:K
**do**
Normalization and move ants
Find neighborhood of current position
Update pheromone and record position $\boldsymbol{\tau}(n)$
**end**
**end**
Update visited Pixel
**end**
**end**

There are two main things in ACO which are needed to calculate construction step and pheromone update [1]. If search clique considered being 4 and 8 then the ant's neighborhood is as shown in Fig 1. (a) and (b) respectively.




(a) (b)

Fig. 1 Ant's neighborhood with search clique (a) 4 (b) 8

The selection of path by ant to move from one position to another is according to the transition Probability Pi,j (n) , It is given by[1].

$$p_{i,j}^{(n)} = \frac{\left(\tau_{i,j}^{(n-1)}\right)^{\alpha}\left(\eta_{i,j}\right)^{\beta}}{\sum_{j\in\Omega_i}\left(\tau_{i,j}^{(n-1)}\right)^{\alpha}\left(\eta_{i,j}\right)^{\beta}}, \quad \text{if } j \in \Omega_i$$

Where $\boldsymbol{\tau_{i,j}^{(n-1)}}$ is the quantity of pheromone on the edge , $\eta_{i,j}$ is the the heuristic information of the edge**,** $\boldsymbol{\Omega i}$ is the neighborhood nodes for the ant given that it is on the node i, the constants $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ represent the influence of pheromone information and heuristic information. The $\sum_{j\in\Omega_i}\left(\tau_{i.j}^{(n-1)}\right)^{\alpha}\left(\eta_{i.j}\right)^{\beta}$
is the normalization factor to limit the values of $\mathbf{Pi,j}^{(n)}$ to [0,1]. The pheromone update follows the equation

$$\tau_{i,j}^{(n)} = (1-\rho)\cdot\tau_{i,j}^{(n-1)} + \sum_{k=1}^{K}\Delta\tau_{i,j}^{(k)}$$

Where $\boldsymbol{\rho\ \varepsilon\ (0,1]}$ is the pheromone evaporation rate, **K** is the no. of ants , $\Delta\tau_{i,j}^{(k)}$ is the quantity of pheromone laid by the $\mathbf{K}^{th}$ ant. The pheromone matrix is updated globally with best solution path availability as

$$\tau_{i,j}^{(n)} = (1-\rho)\cdot\tau_{i,j}^{(n-1)} + \rho\cdot\Delta\tau_{i,j}^{(k_{bs})}$$

Where $\Delta\tau_{i,j}^{(k_{bs})}$ is the amount of pheromone deposited.

## IV. IMAGE EDGE DETECTION USING ACO

The algorithm ACO for Image Edge Detection is summarized as following steps

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### *A. Initialization Process*

In this step ants are dispatched to the image of square size randomly. Initialize the ant's position and Pheromone matrix. Calculate the heuristic matrix $\eta_{i,j}$ as given by the equation[1]

$$\eta_{i,j} = V_c(I_{i,j}) / Z$$

This is also called normalization. $\mathbf{I_{i,j}}$ is the intensity value. Vc is the variation in intensity and Z is the normalization factor where Z = ∑i=mth row∑j=nthcol Vc(Ii,j).

### *B. Construction*

In this step the Ant's are moved as per transition probability as given by [8]

$$p_{i,j}^{(n)} = \frac{\left(\tau_{i,j}^{(n-1)}\right)^{\alpha}\left(\eta_{i,j}\right)^{\beta}}{\sum_{j\in\Omega_i}\left(\tau_{i,j}^{(n-1)}\right)^{\alpha}\left(\eta_{i,j}\right)^{\beta}}, \quad \text{if } j \in \Omega_i$$

*C. Update*

The pheromone matrix is updated as [1]

$$\tau_{i,j}^{(n)} = (1-\rho)\cdot\tau_{i,j}^{(n-1)} + \sum_{k=1}^{K}\Delta\tau_{i,j}^{(k)}$$

*D. Decision*

The decision is made that's if ants are moved or not. The threshold is checked or not. The threshold value [7] is calculated as first Compute mean intensity of image from histogram set

T= mean (I)

Then compute Mean above T (MAT) and Mean below T (MBT) using T.

T (i) = (MAT+MBT)/2

T (i) is then said to be threshold value.

*E. Globally Update*

The pheromone matrix is updated globally with best solution path availability as [1]

$$\tau_{i,j}^{(n)} = (1-\rho)\cdot\tau_{i,j}^{(n-1)} + \rho\cdot\Delta\tau_{i,j}^{(k_{bs})}$$

The flow chart of the above process is shown in Fig 2.

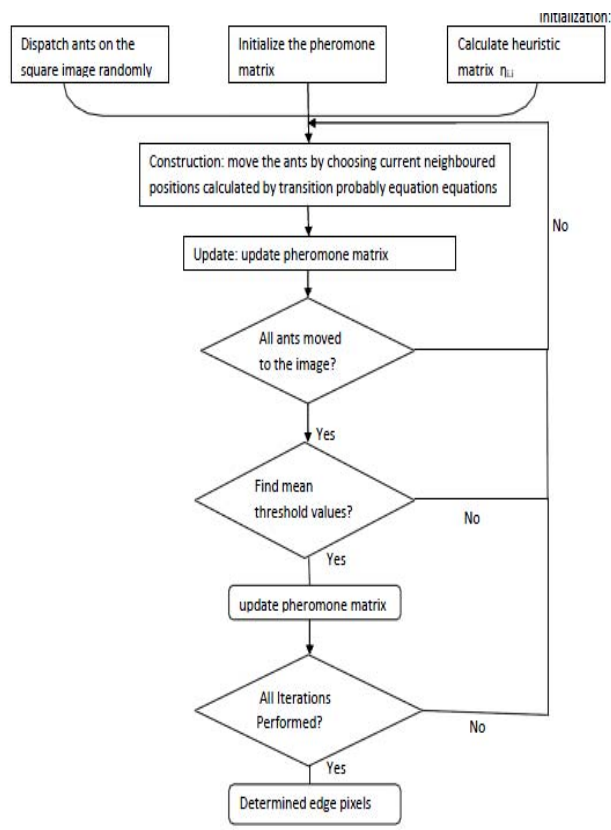


Fig. 2 Flow Chart of new proposed technique

## V. EXPERIMENTAL RESULTS

The Experiment is done on Fig. 3 (a) with resolution 128 X 128 is shown in Fig. 3(b). The values of parameters are taken as:-

N = 10
**α = 1**
**β = 0.1**
**ρ = 0.1**
**ψ = 0.05**
**τ** init = 0.1
**K = 4**
**M= 40**

Fig. 3 (a): A Test image


Fig 3(b): All Optimal Solutions of ACO

## VI. COMPARISON

The results given below in Fig. 3 (c) show the comparison of ACO with other Edge detection Algorithm like Canny and Sobel. PSNR values are calculated and shown in results. PSNR is an engineering term for the ratio between the maximum possible power of a signal and the power of corrupting noise that affects the fidelity of its representation and it should be high for high quality images. PSNR is a rather simple and common way of measuring the loss of information. For good quality of image, the value of PSNR must be high [9].


Fig. 3(c): Comparison of Proposed ACO on the Basis of PSNR Values with other algorithm

Fig. 4: Sample images of resolution 128 X 128 ( 4 (i) - 4 (xx) )

## VII. CONCLUSION

An ACO-based image edge detection approach has been successfully developed. This approach results as superior subjective performance as shown in Fig. 4 and PSNR values of the existing Edge detection Algorithms with suitable parameter values, the algorithm was able to successfully identify edges of the test images. It is clearly shown in graph depicted in Fig. 5 that edge detection using ACO is performing better in comparison to Canny edge and Sobel edge detection.

Table 1: PSNR values of each image using three methods

| | PSNR Values | | |
|---|---|---|---|
| **Sample Image No.** | **ACO Edge Detection** | **Canny Edge Detection** | **Sobel Edge Detection** |
| 4 (i) | 79.8611 | 76.1253 | 76.6577 |
| 4 (ii) | 79.4832 | 76.6577 | 76.8508 |
| 4 (iii) | 79.8611 | 76.6577 | 76.8508 |
| 4 (iv) | 80.1386 | 77.4875 | 77.4875 |
| 4 (v) | 79.1356 | 76.6577 | 76.8508 |
| 4 (vi) | 79.1356 | 76.8508 | 77.058 |
| 4 (vii) | 80.2751 | 77.9705 | 77.9705 |
| 4 (viii) | 78.8137 | 76.2956 | 76.6577 |
| 4 (ix) | 78.5141 | 76.4729 | 76.6577 |
| 4 (x) | 78.5141 | 77.7223 | 77.2547 |
| 4 (xi) | 81.824 | 79.1356 | 79.4832 |
| 4 (xii) | 81.824 | 78.5141 | 78.2338 |
| 4 (xiii) | 79.4832 | 76.2956 | 76.6577 |
| 4(xiv) | 79.8611 | 76.6577 | 77.2647 |
| 4 (xv) | 80.7326 | 78.8137 | 78.8137 |
| 4 (xvi) | 80.275 | 77.9751 | 77.9705 |
| 4 (xvii) | 81.2441 | 77.9705 | 78.5141 |
| 4 (xviii) | 79.1358 | 76.4729 | 76.2956 |
| 4 (ix) | 78.8137 | 76.8508 | 76.6577 |
| 4 (xx) | 77.7223 | 76.2956 | 76.2956 |

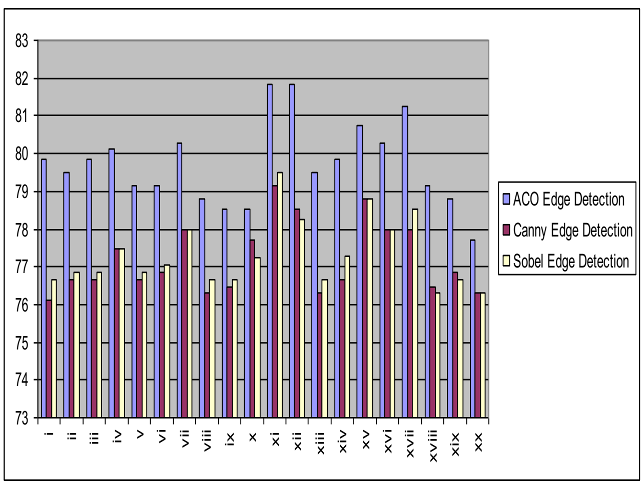


Fig. 5: Comparative Graph of ACO vs. Canny vs. Sobel Edge Detection

## REFERENCES


[1] Anna Veronica, C. Baterina, Carlos M. Oppus," Ant Colony Optimization for Image Edge Detection" Department of Electronics and Communications Engineering Ateneo de Manila University Katipunan Avenue, Loyola Heights, , Quezon City Phillipines.

[2] Jing Tian, Weiyu Yu, and Shengli Xie ," An Ant Colony Optimization Algorithm For Image Edge Detection", 2008 IEEE Congress on Evolutionary Computation (CEC 2008), pp 751-756.

[3] C. Naga Raju, O.Rama Devi, Sharada Mani, Sanam Nagendram, "An Improved Ant Colony Optimization Technique by using Fuzzy Inference Rules for Image Classification and Analysis", IJAEA, Jan 2010,vol I & II, pp 230-234

[4] Alirezae Rezaee ," Extracting Edge of Images with Ant Colony ", Journal of Electrical Engineering, Vol. 59, NO. 1, 2008, pp. 57–59.

[5] Shahab Kamali, Jaroslav Opatrny, "A Position Based Ant Colony Routing Algorithm for Mobile Ad-hoc Networks", Journal of Networks, Vol. 3, No. 4, April 2008, pp 31-41.

[6] M. Dorigo, G. D. Caro, and L. M. Gambardella. Ant algorithms for discrete optimization.Artificial Life, 5(2): pp137–172, 1999.

[7] J. Dr´eo and P. Siarry. Continuous interacting ant colony algorithm based on dense heterarchy. Future Generation Computer Systems, 20(5): pp841–856, June 2004.

[8] O. Cordon, F. Herrera, and T. Stutzle, Special Issue on Ant Colony Optimization: Models and Applications, Mathware and Soft Computing, vol. 9, Dec. 2002

[9] http://fantacci.wikidot.com/licap